

Appendix A: Decomposition Methods

In this appendix, we present the decomposition methods applied to the time series of the variables analyzed in this study, with a focus on the evaluated regions.

Seasonal Decomposition (SD)

Figures A1 to A11 illustrate the application of the SD decomposition method for all variables and regions considered in the study. This method divides the time series into three components: Trend, Seasonality, and Residual.

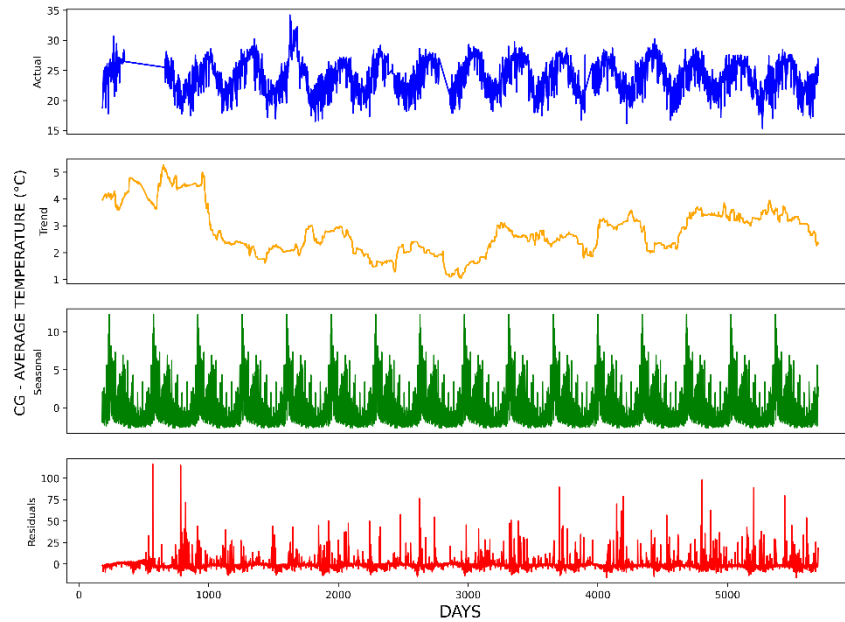


Fig. A1: Components generated from the seasonal decomposition of the Mean Temperature time series for the municipality of Campos dos Goytacazes. Blue – Original Mean Temperature series, Orange – Trend component, Green – Seasonality component, Red – Residual component.

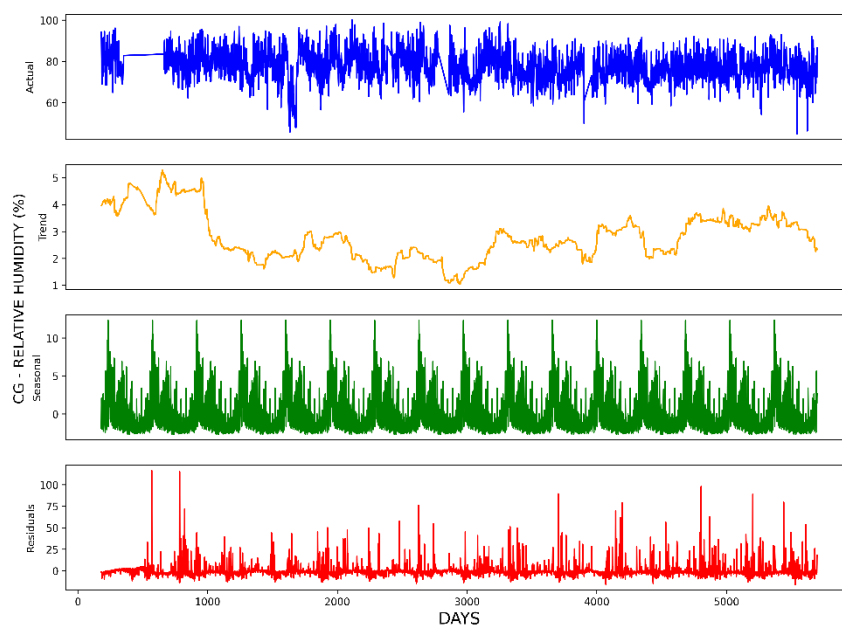


Fig. A2: Components generated from the seasonal decomposition of the Relative Humidity time series for the municipality of Campos dos Goytacazes. Blue – Original Relative Humidity series, Orange – Trend component, Green – Seasonality component, Red – Residual component.

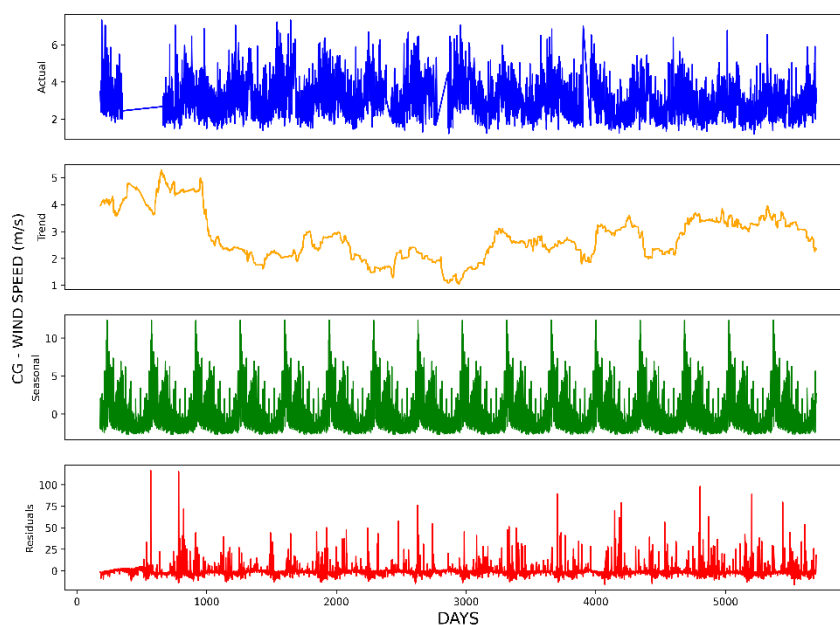


Fig. A3: Components generated from the seasonal decomposition of the Wind Speed time series for the municipality of Campos dos Goytacazes. Blue – Original Wind Speed series, Orange – Trend component, Green – Seasonality component, Red – Residual component.

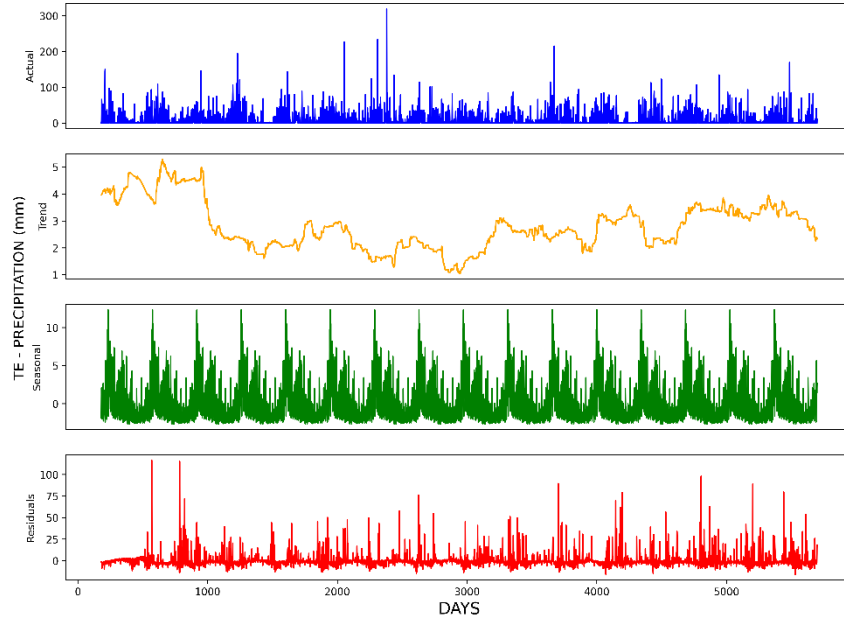


Fig. A4: Components generated from the seasonal decomposition of the Precipitation time series for the municipality of Teresópolis. Blue – Original Precipitation series, Orange – Trend component, Green – Seasonality component, Red – Residual component.

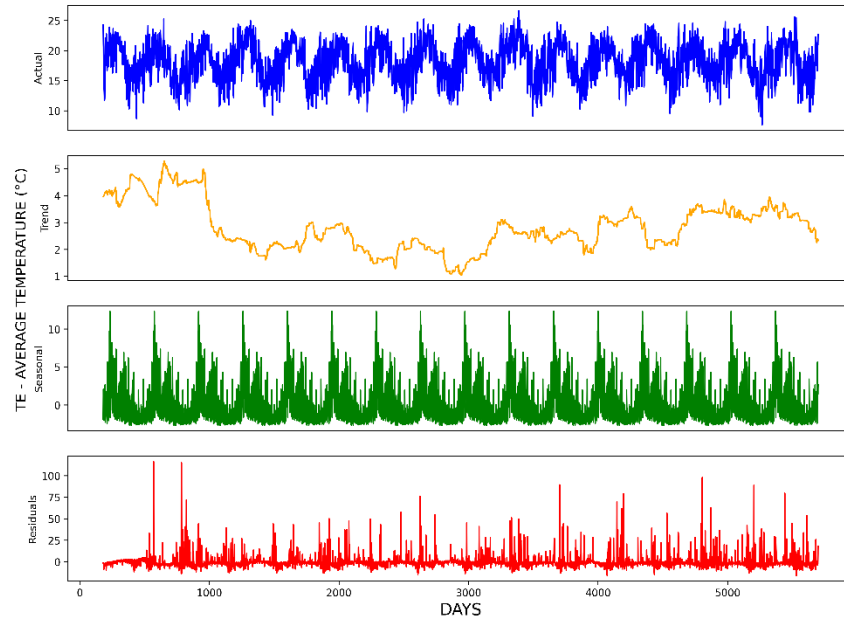


Fig. A5: Components generated from the seasonal decomposition of the Mean Temperature time series for the municipality of Teresópolis. Blue – Original Mean Temperature series, Orange – Trend component, Green – Seasonality component, Red – Residual component.

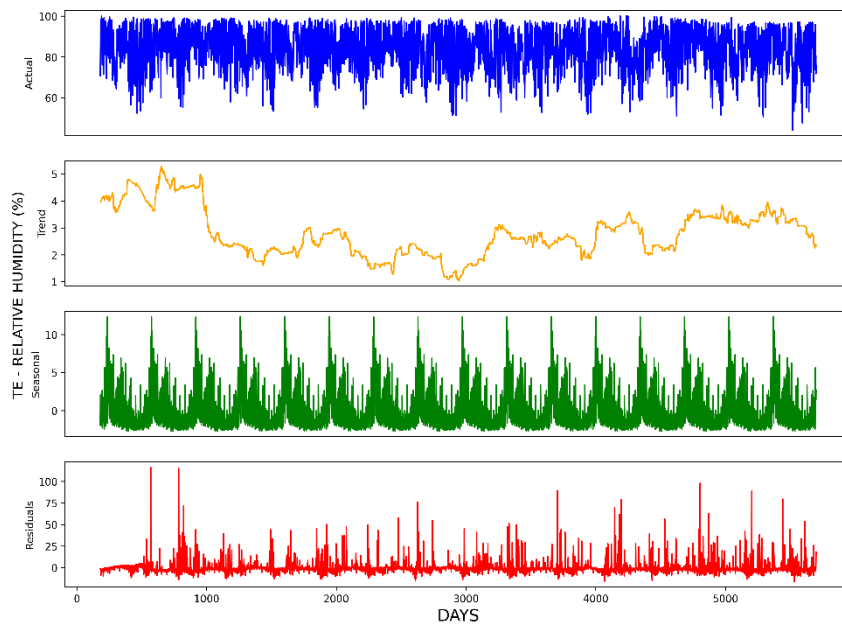


Fig. A6: Components generated from the seasonal decomposition of the Relative Humidity time series for the municipality of Teresópolis. Blue – Original Relative Humidity series, Orange – Trend component, Green – Seasonality component, Red – Residual component.

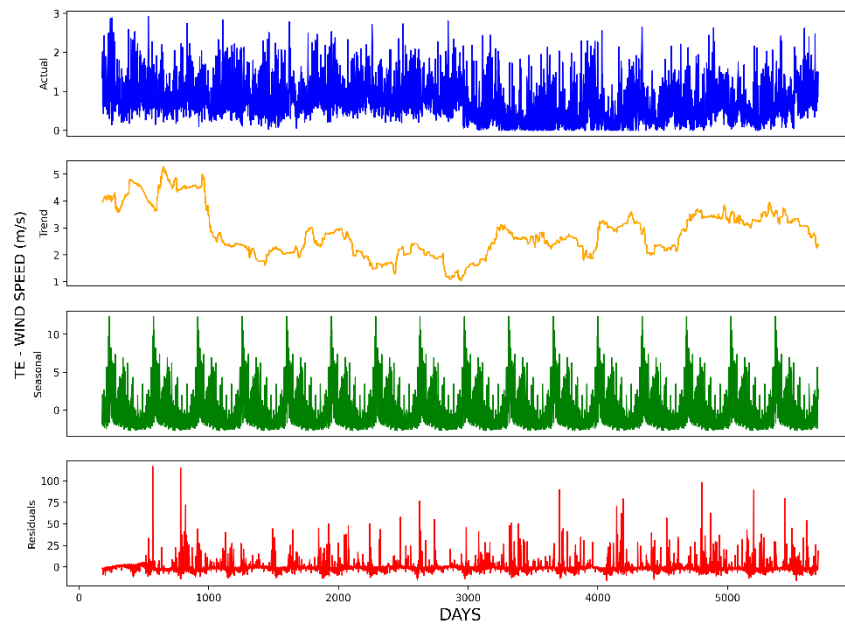


Fig. A7: Components generated from the seasonal decomposition of the Wind Speed time series for the municipality of Teresópolis. Blue – Original Wind Speed series, Orange – Trend component, Green – Seasonality component, Red – Residual component.

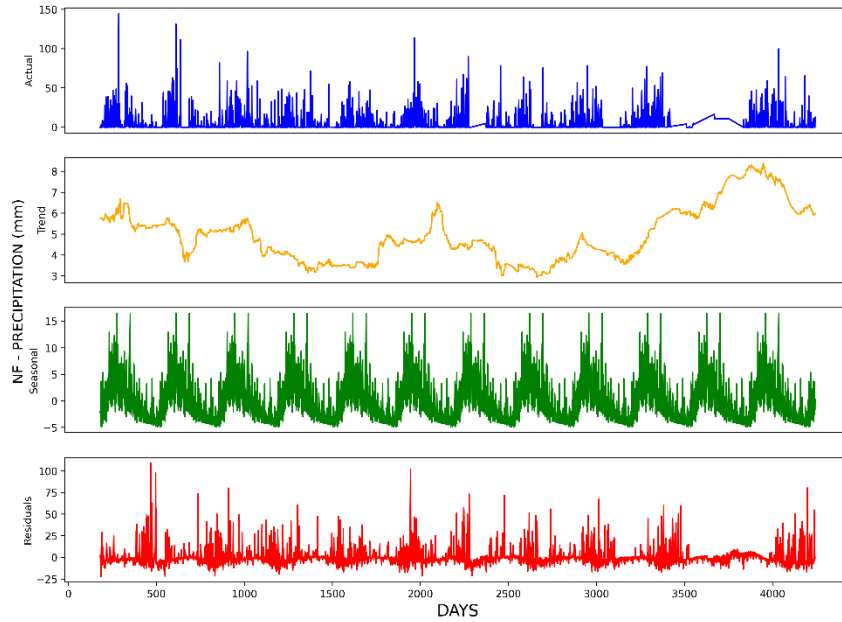


Fig. A8: Components generated from the seasonal decomposition of the Precipitation time series for the municipality of Nova Friburgo. Blue – Original Precipitation series, Orange – Trend component, Green – Seasonality component, Red – Residual component.

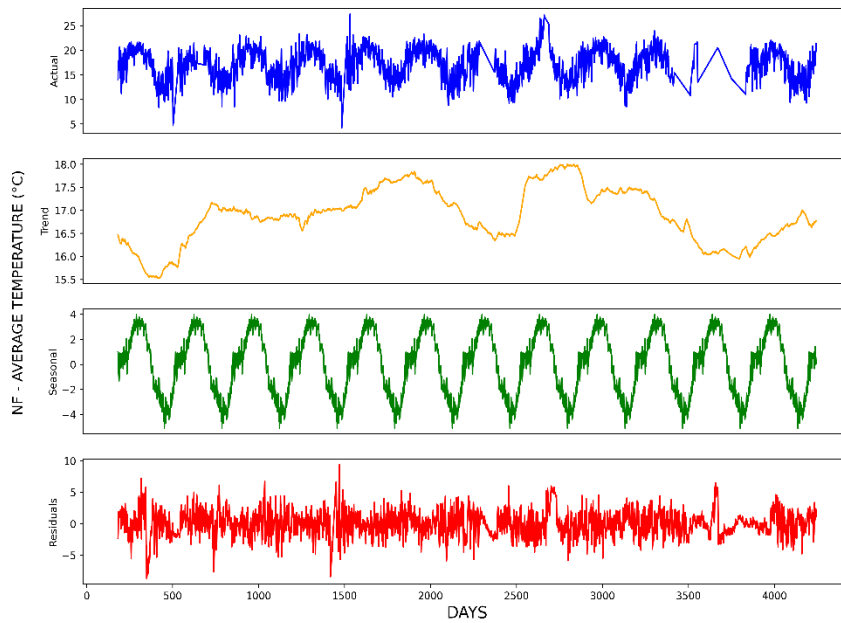


Fig. A9: Components generated from the seasonal decomposition of the Average Temperature time series for the municipality of Nova Friburgo. Blue – Original Average Temperature series, Orange – Trend component, Green – Seasonality component, Red – Residual component.

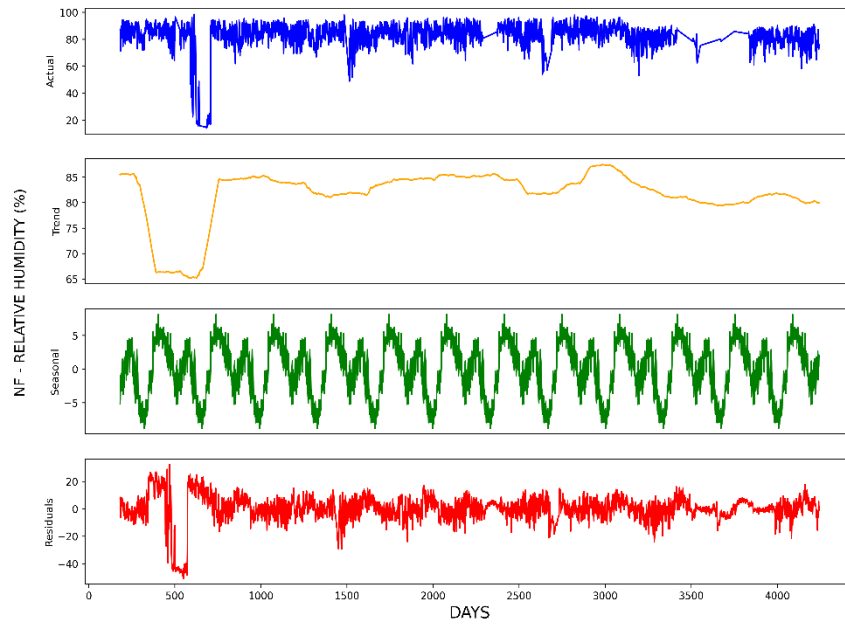


Fig. A10: Components generated from the seasonal decomposition of the Relative Humidity time series for the municipality of Nova Friburgo. Blue – Original Relative Humidity series, Orange – Trend component, Green – Seasonality component, Red – Residual component.



Fig. A11: Components generated from the seasonal decomposition of the Wind Speed time series for the municipality of Nova Friburgo. Blue – Original Wind Speed series, Orange – Trend component, Green – Seasonality component, Red – Residual component.

Ensemble empirical mode decomposition (EEMD)

Figures A12 to A22 illustrate the application of the EEMD decomposition method for all variables and regions considered in the study. This method divides the time series X into multiple IMFs (Intrinsic Mode Functions), each representing distinct frequencies of the original signal.

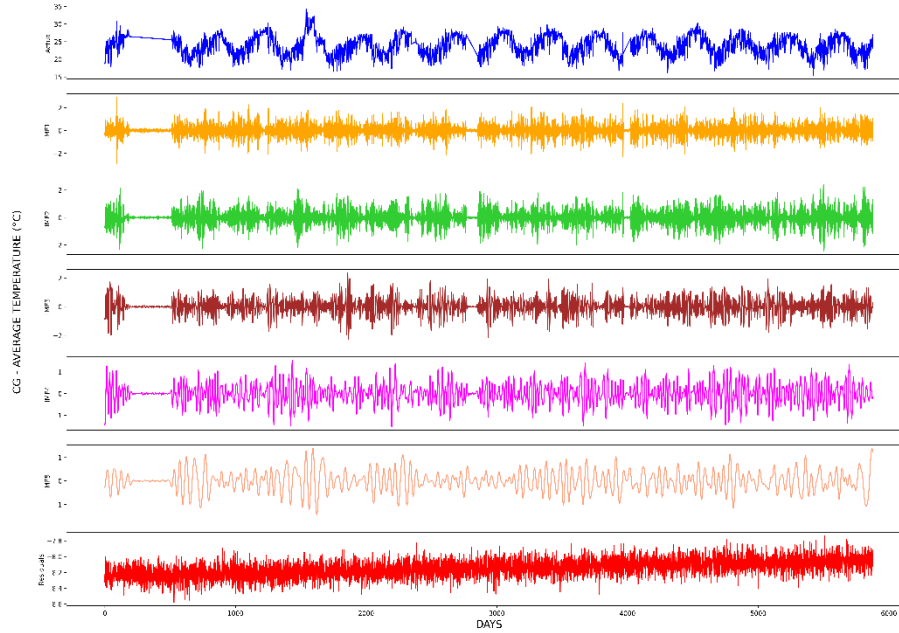


Fig. A12: Components generated from the EEMD decomposition of the average temperature time series for the municipality of Campos dos Goytacazes. Blue – Original average temperature series, Orange, Green, Brown, Magenta, and Salmon – High- and low-frequency components generated by the EEMD decomposition, Red – Residual component.

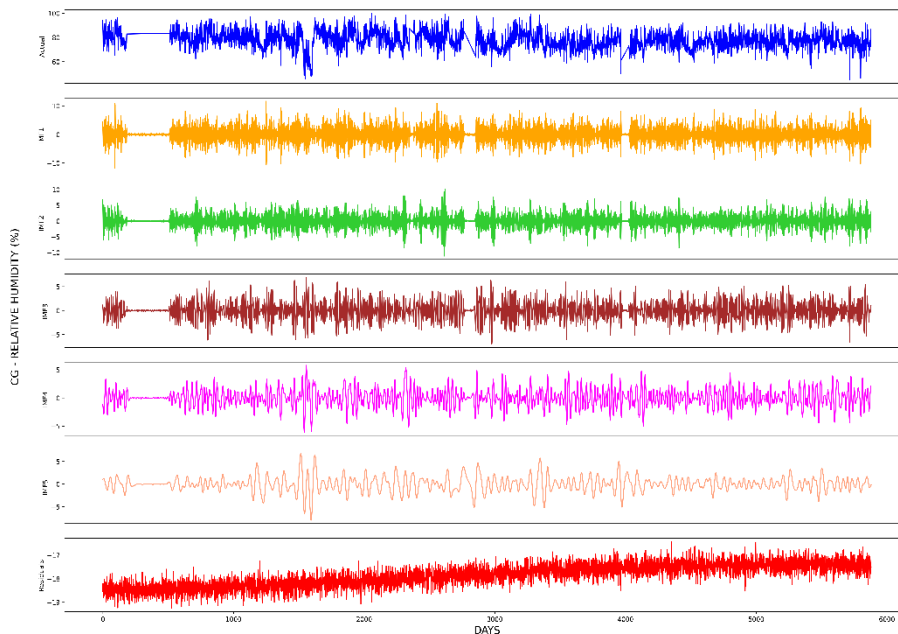


Fig. A13: Components generated from the EEMD decomposition of the Relative Humidity time series for the municipality of Campos dos Goytacazes. Blue – Original Relative Humidity series, Orange, Green, Brown, Magenta, and Salmon – High and low-frequency components generated by the EEMD decomposition, Red – Residual component.

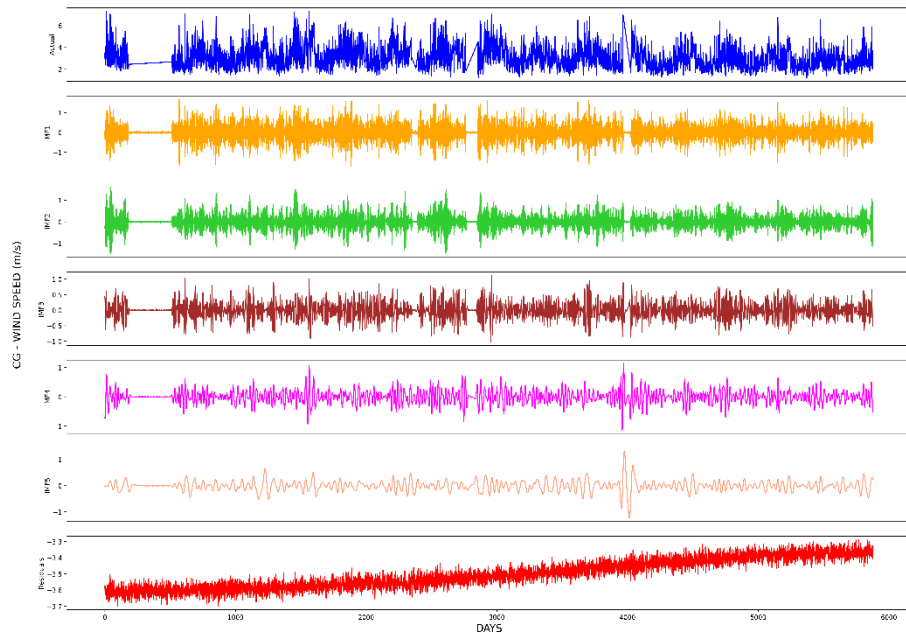


Fig. A14: Components generated from the EEMD decomposition of the Wind Speed time series for the municipality of Campos dos Goytacazes. Blue – Original Wind Speed series, Orange, Green, Brown, Magenta, and Salmon – High and low-frequency components generated by the EEMD decomposition, Red – Residual component.

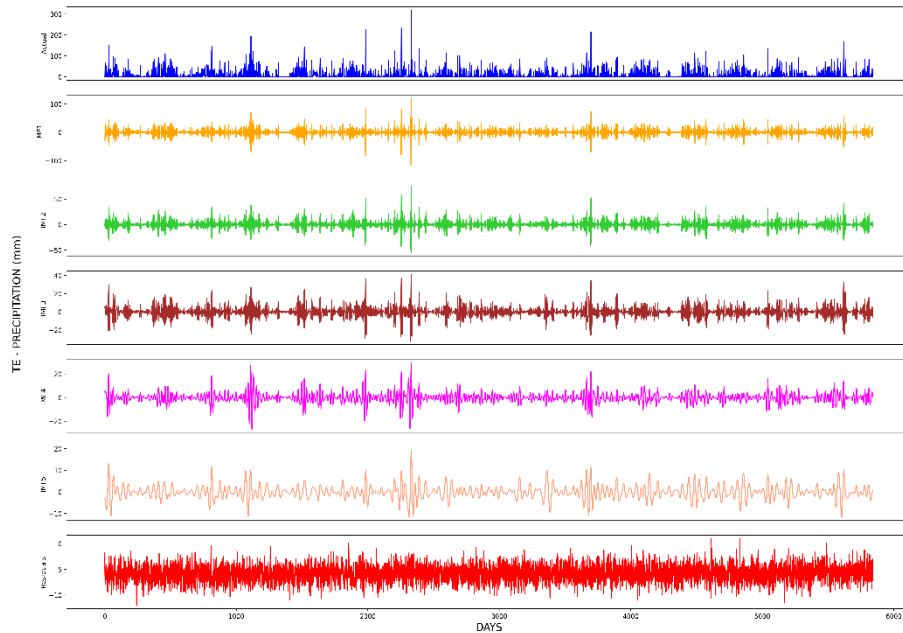


Fig. A15: Components generated from the EEMD decomposition of the precipitation time series for the municipality of Teresópolis. Blue – Original precipitation series, Orange, Green, Brown, Magenta, and Salmon – High and low-frequency components generated by the EEMD decomposition, Red – Residual component.

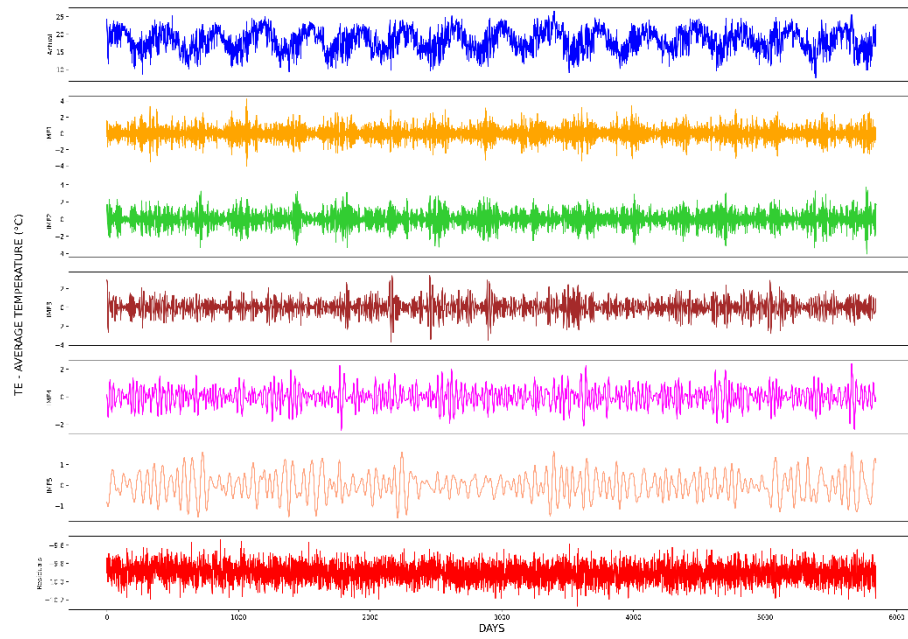


Fig. A16: Components generated from the EEMD decomposition of the average temperature time series for the municipality of Teresópolis. Blue – Original average temperature series, Orange, Green, Brown, Magenta, and Salmon – High and low-frequency components generated by the EEMD decomposition, Red – Residual component.

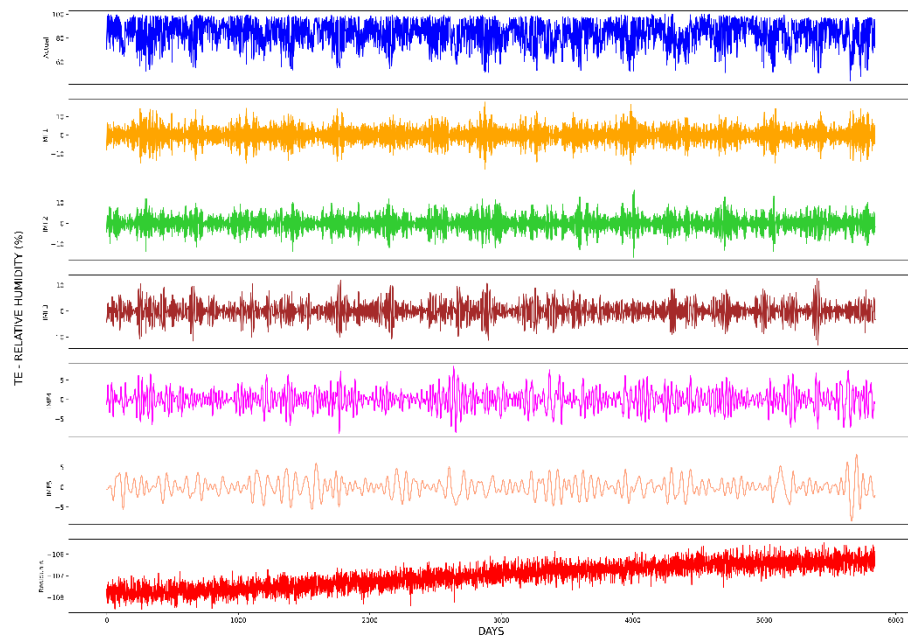


Fig. A17: Components generated from the EEMD decomposition of the relative humidity time series for the municipality of Teresópolis. Blue – Original relative humidity series, Orange, Green, Brown, Magenta, and Salmon – High and low-frequency components generated by the EEMD decomposition, Red – Residual component.

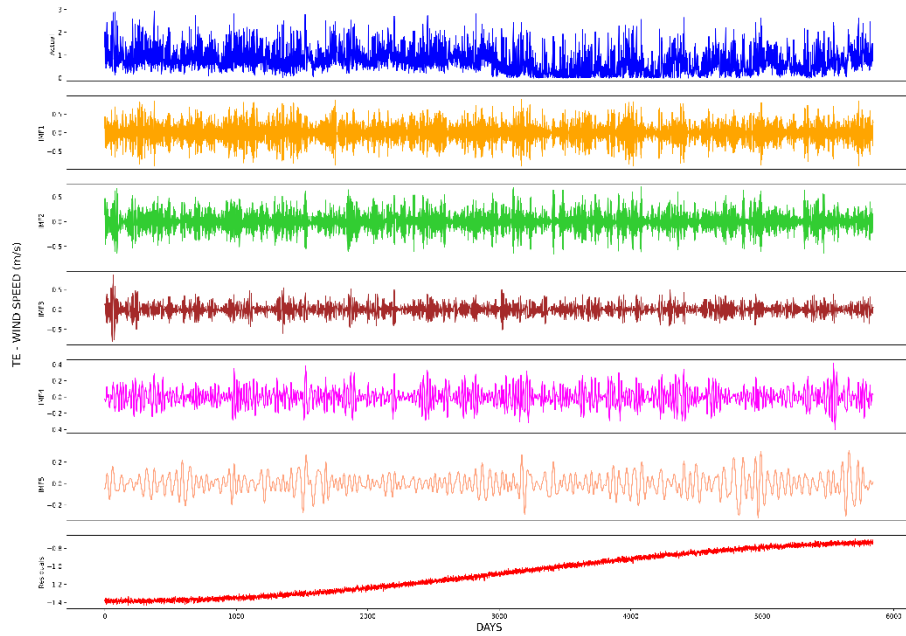


Fig. A18: Components generated from the EEMD decomposition of the wind speed time series for the municipality of Teresópolis. Blue – Original wind speed series, Orange, Green, Brown, Magenta, and Salmon – High and low-frequency components generated by the EEMD decomposition, Red – Residual component.

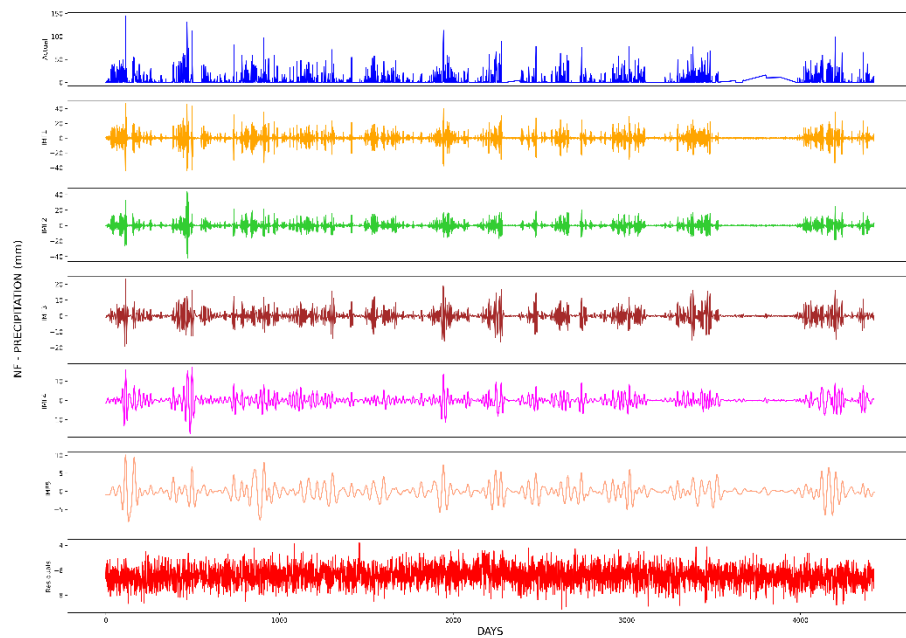


Fig. A19: Components generated from the EEMD decomposition of the Precipitation time series for the municipality of Nova Friburgo. Blue – Original Precipitation series, Orange, Green, Brown, Magenta, and Salmon – High and low-frequency components generated by the EEMD decomposition, Red – Residual component.

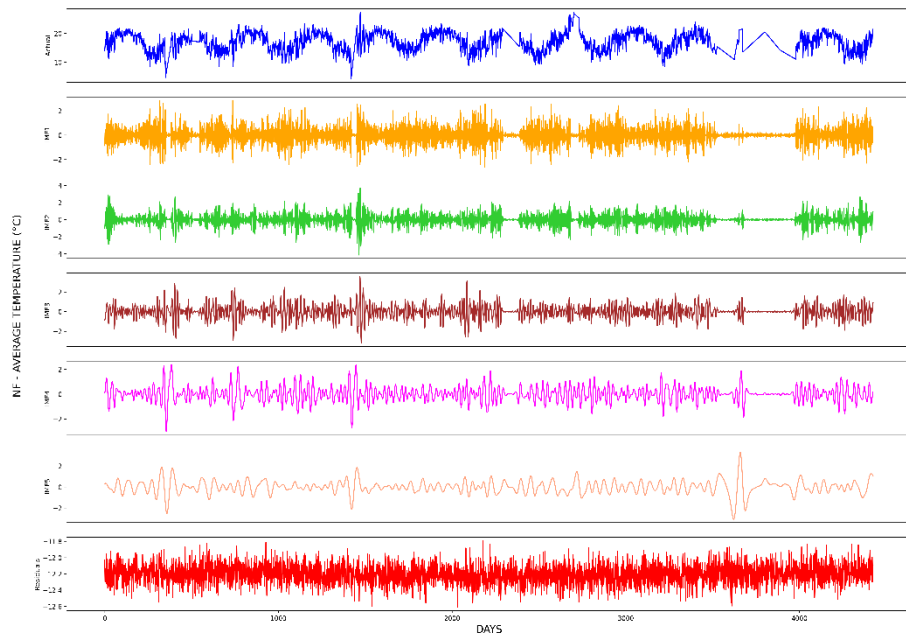


Fig. A20: Components generated from the EEMD decomposition of the average temperature time series for the municipality of Nova Friburgo. Blue – Original average temperature series, Orange, Green, Brown, Magenta, and Salmon – High and low-frequency components generated by the EEMD decomposition, Red – Residual component.

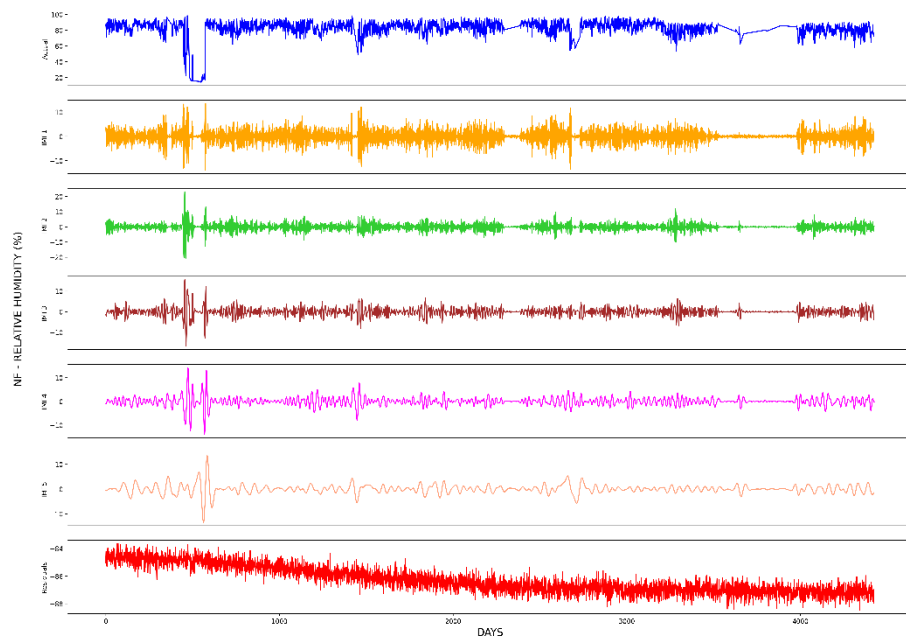


Fig. A21: Components generated from the EEMD decomposition of the relative humidity time series for the municipality of Nova Friburgo. Blue – Original relative humidity series, Orange, Green, Brown, Magenta, and Salmon – High and low-frequency components generated by the EEMD decomposition, Red – Residual component.

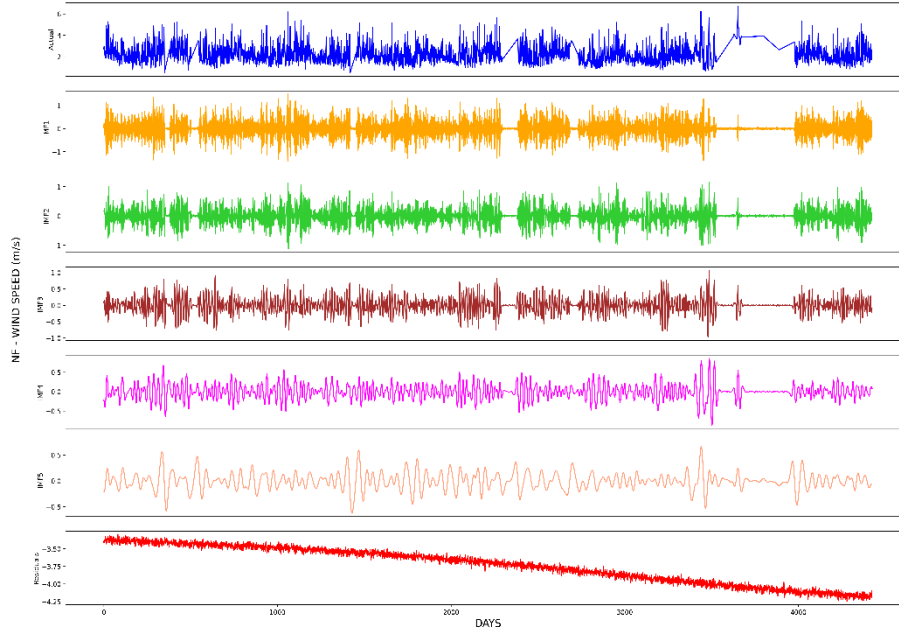


Fig. A22: Components generated from the EEMD decomposition of the wind speed time series for the municipality of Nova Friburgo. Blue – Original wind speed series, Orange, Green, Brown, Magenta, and Salmon – High and low-frequency components generated by the EEMD decomposition, Red – Residual component.

Appendix B: Results

In this appendix, we present the detailed results of the predictions for all the variables analyzed, organized in Tables B1 to B4, using the following metrics: Pearson correlation coefficient (r), coefficient of determination (R^2), mean absolute error (MAE), and root mean squared error (RMSE).

Table B1 Results obtained by the models in forecasting average temperature. Indices: root mean square error (RMSE), mean absolute error (MAE), correlation coefficient (r), coefficient of determination (R^2).

AVERAGE TEMPERATURE (°C)																	
Horizons		1-Step				2-Step				3-Step				4-Step			
Models		RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²
CG	DT	1.20	0.86	0.89	0.79	1.40	1.02	0.84	0.72	1.51	1.12	0.81	0.67	1.57	1.17	0.79	0.63
	RF	1.16	0.83	0.89	0.80	1.36	0.98	0.85	0.73	1.48	1.09	0.82	0.68	1.54	1.14	0.80	0.65
	CNN-LSTM	1.16	0.83	0.90	0.81	1.36	1.03	0.86	0.73	1.47	1.09	0.82	0.69	1.52	1.13	0.81	0.65
	EEMD-DT	0.87	0.65	0.94	0.89	1.12	0.83	0.90	0.82	1.23	0.91	0.88	0.78	1.28	0.96	0.87	0.76
	EEMD-RF	0.72	0.52	0.96	0.92	0.94	0.67	0.93	0.87	1.07	0.78	0.91	0.83	1.14	0.84	0.90	0.81
	EEMD-CNN-LSTM	0.41	0.30	0.98	0.97	0.57	0.41	0.97	0.95	0.71	0.53	0.96	0.92	0.83	0.62	0.94	0.89
	SD-DT	0.61	0.45	0.97	0.94	0.98	0.72	0.92	0.85	1.22	0.91	0.87	0.76	1.36	1.01	0.83	0.71
	SD-RF	0.45	0.30	0.98	0.97	0.84	0.60	0.93	0.88	1.10	0.80	0.89	0.80	1.25	0.93	0.85	0.74
	SD-CNN-LSTM	0.30	0.23	0.99	0.98	0.68	0.52	0.95	0.90	0.95	0.70	0.90	0.82	1.15	0.87	0.87	0.76
Models		RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²
TE	DT	1.70	1.27	0.83	0.70	1.98	1.52	0.76	0.58	2.11	1.65	0.70	0.50	2.23	1.75	0.67	0.45
	RF	1.65	1.23	0.84	0.71	1.95	1.49	0.76	0.59	2.08	1.63	0.71	0.52	2.19	1.72	0.68	0.47
	CNN-LSTM	1.63	1.22	0.85	0.72	1.97	1.49	0.77	0.58	2.08	1.64	0.71	0.51	2.17	1.70	0.68	0.48
	EEMD-DT	1.20	0.91	0.92	0.85	1.57	1.21	0.85	0.73	1.74	1.36	0.81	0.66	1.82	1.42	0.79	0.63
	EEMD -RF	1.02	0.73	0.94	0.89	1.33	1.01	0.90	0.81	1.53	1.19	0.86	0.74	1.65	1.29	0.83	0.70
	EEMD-CNN-LSTM	0.61	0.44	0.98	0.96	0.83	0.61	0.96	0.92	1.06	0.79	0.93	0.86	1.23	0.93	0.91	0.82
	SD-DT	0.84	0.64	0.96	0.92	1.32	1.02	0.89	0.79	1.69	1.32	0.80	0.65	1.89	1.48	0.75	0.58
	SD-RF	0.61	0.41	0.98	0.96	1.14	0.86	0.91	0.83	1.53	1.18	0.82	0.70	1.76	1.38	0.77	0.61
	SD-CNN-LSTM	0.10	0.08	0.99	0.99	0.87	0.68	0.93	0.86	1.33	1.03	0.84	0.72	1.60	1.25	0.78	0.64
Models		RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²
NF	DT	1.50	1.09	0.88	0.78	1.73	1.27	0.84	0.71	1.84	1.36	0.81	0.66	1.89	1.41	0.80	0.65
	RF	1.46	1.05	0.89	0.80	1.68	1.22	0.85	0.73	1.79	1.31	0.82	0.68	1.86	1.37	0.81	0.66
	CNN-LSTM	1.42	1.04	0.90	0.81	1.68	1.23	0.86	0.73	1.79	1.34	0.83	0.68	1.84	1.39	0.81	0.67
	EEMD-DT	1.20	0.87	0.93	0.86	1.44	1.06	0.89	0.80	1.52	1.13	0.88	0.77	1.60	1.19	0.86	0.75
	EEMD -RF	0.99	0.68	0.95	0.90	1.25	0.89	0.92	0.85	1.34	0.97	0.90	0.82	1.42	1.03	0.89	0.80
	EEMD-CNN-LSTM	0.48	0.35	0.98	0.97	0.76	0.56	0.97	0.94	0.86	0.64	0.96	0.92	0.97	0.72	0.94	0.90
	SD-DT	0.88	0.63	0.96	0.92	1.26	0.92	0.91	0.84	1.50	1.11	0.87	0.77	1.65	1.22	0.84	0.72
	SD-RF	0.68	0.42	0.97	0.95	1.11	0.77	0.93	0.87	1.35	0.97	0.89	0.80	1.52	1.09	0.86	0.75
	SD-CNN-LSTM	0.28	0.26	0.99	0.99	0.77	0.55	0.95	0.90	1.19	0.90	0.91	0.82	1.50	1.13	0.87	0.76

Table B2 Results obtained by the models in wind speed forecasting. Indices: root mean square error (*RMSE*), mean absolute error (*MAE*), correlation coefficient (*r*), coefficient of determination (*R*²).

WIND SPEED (m/s)																	
Horizons		1-Step				2-Step				3-Step				4-Step			
Models		RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²
CG	DT	0.82	0.62	0.56	0.31	0.87	0.66	0.46	0.22	0.87	0.66	0.45	0.21	0.89	0.68	0.40	0.17
	RF	0.80	0.60	0.59	0.35	0.85	0.64	0.50	0.25	0.85	0.65	0.48	0.24	0.87	0.66	0.45	0.21
	CNN-LSTM	0.79	0.61	0.60	0.36	0.84	0.65	0.50	0.24	0.85	0.65	0.48	0.24	0.88	0.67	0.44	0.19
	EEMD-DT	0.57	0.42	0.82	0.67	0.69	0.51	0.71	0.50	0.73	0.54	0.66	0.44	0.75	0.56	0.64	0.41
	EEMD -RF	0.47	0.34	0.88	0.77	0.60	0.44	0.78	0.62	0.66	0.49	0.73	0.54	0.69	0.52	0.70	0.50
	EEMD-CNN-LSTM	0.38	0.26	0.92	0.85	0.47	0.34	0.87	0.75	0.52	0.38	0.83	0.69	0.58	0.43	0.79	0.63
	SD-DT	0.37	0.27	0.92	0.85	0.60	0.44	0.74	0.58	0.71	0.54	0.63	0.43	0.76	0.57	0.57	0.36
	SD-RF	0.26	0.16	0.96	0.92	0.53	0.38	0.78	0.64	0.64	0.48	0.67	0.49	0.71	0.53	0.61	0.41
	SD-CNN-LSTM	0.12	0.11	0.99	0.98	0.41	0.31	0.81	0.69	0.57	0.44	0.69	0.52	0.65	0.50	0.63	0.44
Models		RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²
TE	DT	0.44	0.33	0.49	0.23	0.45	0.34	0.43	0.19	0.46	0.36	0.39	0.15	0.46	0.36	0.40	0.16
	RF	0.43	0.33	0.52	0.26	0.45	0.34	0.45	0.21	0.46	0.35	0.41	0.17	0.46	0.35	0.42	0.18
	CNN-LSTM	0.42	0.32	0.55	0.30	0.45	0.34	0.46	0.22	0.46	0.35	0.42	0.18	0.46	0.35	0.42	0.18
	EEMD-DT	0.33	0.24	0.75	0.56	0.36	0.27	0.68	0.47	0.39	0.29	0.62	0.39	0.41	0.31	0.57	0.33
	EEMD -RF	0.28	0.20	0.83	0.69	0.33	0.24	0.76	0.57	0.36	0.27	0.70	0.48	0.38	0.29	0.66	0.43
	EEMD-CNN-LSTM	0.21	0.14	0.90	0.81	0.27	0.19	0.83	0.69	0.29	0.21	0.80	0.64	0.24	0.24	0.74	0.56
	SD-DT	0.20	0.14	0.91	0.83	0.32	0.24	0.71	0.54	0.38	0.29	0.57	0.38	0.41	0.31	0.51	0.31
	SD-RF	0.14	0.09	0.96	0.91	0.28	0.21	0.76	0.61	0.35	0.27	0.61	0.43	0.38	0.29	0.55	0.36
	SD-CNN-LSTM	0.01	0.01	0.99	0.99	0.22	0.17	0.78	0.66	0.31	0.24	0.64	0.48	0.35	0.27	0.58	0.39
Models		RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²
NF	DT	0.67	0.48	0.61	0.37	0.46	0.35	0.42	0.18	0.71	0.52	0.52	0.27	0.72	0.53	0.50	0.25
	RF	0.66	0.47	0.63	0.39	0.45	0.34	0.45	0.20	0.70	0.51	0.54	0.29	0.70	0.51	0.53	0.28
	CNN-LSTM	0.65	0.47	0.64	0.40	0.45	0.34	0.46	0.22	0.71	0.52	0.54	0.28	0.71	0.51	0.53	0.27
	EEMD-DT	0.51	0.36	0.80	0.63	0.57	0.41	0.72	0.53	0.61	0.44	0.68	0.46	0.63	0.46	0.65	0.42
	EEMD -RF	0.41	0.28	0.87	0.76	0.51	0.36	0.79	0.63	0.55	0.39	0.75	0.56	0.57	0.41	0.72	0.52
	EEMD-CNN-LSTM	0.29	0.21	0.93	0.87	0.40	0.29	0.88	0.76	0.44	0.32	0.84	0.70	0.47	0.34	0.81	0.65
	SD-DT	0.33	0.23	0.92	0.84	0.52	0.37	0.74	0.58	0.58	0.42	0.67	0.47	0.63	0.46	0.60	0.39
	SD-RF	0.23	0.14	0.96	0.92	0.46	0.32	0.78	0.64	0.54	0.38	0.70	0.52	0.59	0.42	0.65	0.45
	SD-CNN-LSTM	0.08	0.07	0.99	0.98	0.39	0.29	0.82	0.69	0.50	0.37	0.71	0.54	0.54	0.39	0.66	0.48

Table B3 Results obtained by the models in relative humidity forecasting. Indices: root mean square error (*RMSE*), mean absolute error (*MAE*), correlation coefficient (*r*), coefficient of determination (*R*²).

RELATIVE HUMIDITY (%)																	
Horizons		1-Step				2-Step				3-Step				4-Step			
Models		RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²
CG	DT	5.15	3.79	0.67	0.46	5.58	4.15	0.60	0.37	5.71	4.27	0.57	0.33	5.83	4.39	0.55	0.31
	RF	5.02	3.71	0.69	0.48	5.47	4.08	0.62	0.39	5.63	4.23	0.59	0.35	5.77	4.33	0.56	0.32
	CNN-LSTM	5.02	3.77	0.70	0.49	5.45	4.08	0.63	0.40	5.71	4.35	0.59	0.33	5.79	4.43	0.56	0.32
	EEMD-DT	3.75	2.77	0.84	0.71	4.49	3.31	0.76	0.59	4.68	3.49	0.74	0.55	5.07	3.81	0.69	0.47
	EEMD -RF	3.16	2.21	0.89	0.79	3.96	2.88	0.82	0.68	4.26	3.14	0.79	0.62	4.63	3.44	0.75	0.56
	EEMD-CNN-LSTM	2.30	1.61	0.94	0.89	2.95	2.19	0.90	0.81	3.31	2.48	0.88	0.76	3.70	2.78	0.84	0.70
	SD-DT	2.55	1.82	0.93	0.86	3.91	2.90	0.80	0.65	4.61	3.43	0.71	0.52	5.01	3.76	0.65	0.45
	SD-RF	1.90	1.18	0.96	0.92	3.55	2.55	0.82	0.70	4.33	3.16	0.73	0.56	4.77	3.52	0.68	0.49
SD-CNN-LSTM	0.62	0.55	0.99	0.99	3.08	2.40	0.85	0.73	3.81	2.92	0.76	0.59	4.60	3.55	0.68	0.50	
Models		RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²
TE	DT	7.53	5.74	0.61	0.37	8.39	6.67	0.47	0.24	8.59	6.89	0.40	0.18	8.86	7.12	0.35	0.15
	RF	7.38	5.63	0.63	0.39	8.28	6.57	0.49	0.26	8.51	6.81	0.42	0.20	8.79	7.07	0.37	0.16
	CNN-LSTM	7.32	5.60	0.63	0.40	8.26	6.56	0.50	0.26	8.53	6.88	0.42	0.20	8.78	7.08	0.38	0.16
	EEMD-DT	5.48	4.09	0.81	0.66	6.48	5.03	0.74	0.54	7.09	5.50	0.66	0.44	7.69	5.96	0.60	0.36
	EEMD -RF	4.70	3.43	0.87	0.75	5.62	4.30	0.82	0.66	6.22	4.84	0.76	0.57	6.98	5.47	0.69	0.47
	EEMD-CNN-LSTM	3.49	2.51	0.93	0.86	4.22	3.12	0.89	0.80	4.91	3.69	0.85	0.72	5.50	4.17	0.80	0.65
	SD-DT	3.40	2.53	0.93	0.87	5.53	4.26	0.78	0.63	6.85	5.37	0.61	0.43	7.54	5.99	0.52	0.34
	SD-RF	2.27	1.51	0.97	0.94	4.94	3.78	0.83	0.68	6.39	5.03	0.64	0.48	7.13	5.67	0.55	0.38
SD-CNN-LSTM	0.38	0.34	0.99	0.99	3.92	3.13	0.84	0.73	5.69	4.51	0.67	0.51	6.62	5.34	0.57	0.40	
Models		RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²
NF	DT	5.49	3.76	0.91	0.83	5.90	4.06	0.90	0.81	6.20	4.28	0.88	0.78	6.21	4.30	0.88	0.79
	RF	4.95	3.42	0.93	0.86	5.51	3.84	0.91	0.83	5.86	4.06	0.90	0.81	6.09	4.16	0.89	0.79
	CNN-LSTM	4.87	3.46	0.93	0.87	5.61	4.07	0.91	0.83	5.85	4.13	0.90	0.81	6.16	4.33	0.89	0.79
	EEMD-DT	4.52	2.93	0.94	0.89	5.16	3.49	0.92	0.85	5.36	3.61	0.91	0.84	5.41	3.67	0.91	0.83
	EEMD -RF	3.34	2.17	0.97	0.94	4.13	2.76	0.95	0.90	4.65	3.08	0.93	0.88	4.95	3.29	0.93	0.86
	EEMD-CNN-LSTM	2.44	1.83	0.98	0.96	3.23	2.26	0.97	0.94	4.18	2.80	0.94	0.90	4.16	2.77	0.95	0.90
	SD-DT	2.93	1.96	0.97	0.95	4.27	2.91	0.94	0.89	4.97	3.36	0.92	0.85	5.55	3.73	0.90	0.81
	SD-RF	2.23	1.26	0.98	0.97	3.80	2.47	0.95	0.90	4.65	3.07	0.93	0.86	5.23	3.45	0.91	0.83
SD-CNN-LSTM	1.50	1.32	0.99	0.98	2.88	2.03	0.95	0.90	4.22	2.94	0.93	0.88	5.02	3.56	0.92	0.84	

Table B4 Results obtained by the models in precipitation forecasting. Indices: root mean square error (RMSE), mean absolute error (MAE), correlation coefficient (r), coefficient of determination (R^2).

PRECIPITATION (mm)																	
Horizons		1-Step				2-Step				3-Step				4-Step			
Models		RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²
CG	DT	7.68	3.64	0.31	0.09	8.39	3.80	0.22	0.05	8.33	3.84	0.20	0.04	8.54	3.90	0.16	0.02
	RF	7.67	3.63	0.32	0.10	8.40	3.80	0.23	0.05	8.33	3.82	0.20	0.04	8.52	3.89	0.17	0.03
	CNN-LSTM	7.72	3.63	0.30	0.09	8.43	4.01	0.21	0.05	8.37	3.78	0.18	0.03	8.55	4.15	0.16	0.02
	EEMD-DT	4.52	1.80	0.83	0.68	6.24	2.36	0.68	0.47	6.63	2.57	0.62	0.39	6.75	2.79	0.60	0.38
	EEMD -RF	4.05	1.56	0.86	0.74	5.82	2.14	0.73	0.54	6.13	2.39	0.69	0.48	6.41	2.64	0.65	0.43
	EEMD-CNN-LSTM	3.45	1.52	0.90	0.81	4.62	1.77	0.84	0.71	5.20	2.17	0.78	0.61	6.01	2.48	0.70	0.48
	SD-DT	3.74	1.16	0.88	0.78	6.86	2.59	0.52	0.34	7.43	3.12	0.39	0.22	7.68	3.31	0.32	0.18
	SD-RF	3.14	0.86	0.92	0.84	6.17	2.30	0.57	0.43	6.97	2.92	0.43	0.29	7.32	3.19	0.37	0.23
	SD-CNN-LSTM	0.29	0.24	0.99	0.99	4.80	2.05	0.60	0.51	6.05	2.81	0.47	0.36	6.64	3.14	0.38	0.26
Models		RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²
TE	DT	15.08	8.58	0.32	0.10	16.43	9.36	0.21	0.05	16.06	9.53	0.15	0.03	16.87	9.67	0.14	0.02
	RF	15.01	8.55	0.33	0.11	16.36	9.30	0.23	0.06	16.03	9.53	0.16	0.03	16.84	9.70	0.14	0.02
	CNN-LSTM	15.05	8.66	0.33	0.10	16.42	9.77	0.22	0.05	15.98	9.37	0.17	0.04	16.81	9.41	0.15	0.03
	EEMD-DT	9.45	4.74	0.80	0.64	12.66	6.25	0.64	0.42	12.58	6.65	0.63	0.36	13.76	7.24	0.57	0.34
	EEMD -RF	7.74	3.90	0.87	0.76	10.69	5.50	0.76	0.58	11.41	6.10	0.70	0.47	12.75	6.88	0.65	0.43
	EEMD-CNN-LSTM	5.85	3.16	0.93	0.86	9.51	5.26	0.81	0.67	9.62	4.79	0.77	0.61	11.25	5.95	0.70	0.53
	SD-DT	6.62	2.63	0.90	0.82	12.29	6.11	0.56	0.41	13.22	7.31	0.41	0.27	14.94	8.18	0.31	0.20
	SD-RF	4.73	1.65	0.95	0.91	10.98	5.50	0.60	0.48	12.51	6.95	0.43	0.31	14.34	7.89	0.35	0.24
	SD-CNN-LSTM	1.62	1.14	0.99	0.98	9.10	5.18	0.64	0.54	11.27	6.56	0.44	0.35	13.60	7.75	0.36	0.26
Models		RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²	RMSE	MAE	r	R ²
NF	DT	10.36	5.64	0.31	0.09	10.71	5.93	0.22	0.05	10.69	5.83	0.23	0.05	10.65	5.90	0.22	0.05
	RF	10.42	5.53	0.30	0.08	10.68	5.84	0.24	0.05	10.68	5.77	0.23	0.05	10.64	5.82	0.22	0.05
	CNN-LSTM	10.79	5.58	0.22	0.01	10.82	6.15	0.21	0.03	10.80	5.91	0.19	0.03	10.67	6.08	0.21	0.04
	EEMD-DT	6.47	2.95	0.80	0.64	7.77	3.73	0.69	0.48	8.32	3.86	0.64	0.42	8.70	4.29	0.59	0.35
	EEMD -RF	5.84	2.64	0.84	0.71	6.77	3.30	0.78	0.61	7.73	3.65	0.70	0.49	8.11	3.99	0.65	0.43
	EEMD-CNN-LSTM	5.19	2.29	0.88	0.77	6.01	2.81	0.83	0.69	6.84	3.30	0.77	0.60	7.59	4.25	0.70	0.49
	SD-DT	5.75	2.09	0.84	0.72	8.05	3.96	0.57	0.40	9.15	4.66	0.44	0.27	9.62	5.17	0.36	0.20
	SD-RF	4.12	1.37	0.92	0.85	7.21	3.57	0.63	0.47	8.54	4.30	0.49	0.33	9.10	4.79	0.41	0.25
	SD-CNN-LSTM	0.54	0.44	0.99	0.99	5.64	2.85	0.64	0.52	7.77	4.59	0.51	0.37	8.24	4.53	0.43	0.29